# BASE CHANGE OF THE NAIVE COTANGENT COMPLEX OF A PRESENTATION IS AN ISOMORPHISM ON H^0 AND ONTO ON H^{-1}

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 22770-22786 (canonical tag 0FUZ) and the counted proof is the proof environment at source lines 22788-22807, which immediately follows it and is its own proof.

*Editorial note.* The macro NL of the naive cotangent complex is supplied by the retained preamble, which copies the shared Stacks preamble verbatim; the displayed square of the proof is the author's own Xy-pic markup.

*Editorial note.* The proof transfers the presentation-level statement to the canonical complexes by using that the two are homotopic, a fact the author invokes without a label; the transfer is the last step and is preserved verbatim, while the substantive comparison is carried out for a polynomial presentation inside the counted span.

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*Allowed standard prerequisites (editorial).*

- The naive cotangent complex of a presentation P -> S with kernel I is the two-term complex I/I^2 -> S tensor over P for the module of differentials, displayed in degrees -1 and 0. used at proof lines 22789-22800, where I and I' are introduced and the displayed square is identified with NL(alpha) tensor S' on the left and NL(alpha') on the right. Fixes the explicit two-term complexes whose base change map has to be analysed.
- Tensor product is right exact, so the natural map I tensor_R R' -> I' is surjective when I' is defined as the kernel of P tensor R' -> S tensor R'. used at proof lines 22801-22803. Makes the bottom horizontal arrow of the displayed square surjective, which is the assertion about H^{-1}.

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- The naive cotangent complex of a ring map is independent of the chosen presentation up to homotopy, so the two complexes in the second displayed map are homotopic to the complexes of a fixed presentation. invoked at proof lines 22804-22806, where the statement for NL_{S/R} is deduced because the complexes are homotopic to NL(alpha) tensor S' and to NL(alpha'). Transfers the presentation-level statement to the canonical naive cotangent complex of the ring map, which is the second assertion of the lemma.

**Lemma 0.2.** *Let $R \to S$ and $R \to R'$ be ring maps. Let $\alpha : P \to S$ be a presentation of $S$ over $R$. Then $\alpha' : P \otimes_R R' \to S \otimes_R R'$ is a presentation of $S' = S \otimes_R R'$ over $R'$. The canonical map*

$$NL(\alpha) \otimes_S S' \to NL(\alpha')$$

*is an isomorphism on $H^0$ and surjective on $H^{-1}$. In particular, the canonical map*

$$NL_{S/R} \otimes_S S' \to NL_{S'/R'}$$

*is an isomorphism on $H^0$ and surjective on $H^{-1}$.*

*Proof.* Denote $I = \mathrm{Ker}(P \to S)$. Denote $P' = P \otimes_R R'$ and $I' = \mathrm{Ker}(P' \to S')$. Suppose $P$ is a polynomial algebra on $x_j$ for $j \in J$. The map displayed in the lemma becomes

$$\begin{array}{ccc} \bigoplus_{j \in J} S' \mathrm{d}x_j & \longrightarrow & \bigoplus_{j \in J} S' \mathrm{d}x_j \\ \uparrow & & \uparrow \\ I/I^2 \otimes_S S' & \longrightarrow & I'/(I')^2 \end{array}$$

where the left column is $NL(\alpha) \otimes_S S'$ and the right column is $NL(\alpha')$. By right exactness of tensor product we see that $I \otimes_R R' \to I'$ is surjective. Hence the bottom arrow is a surjection. This proves the first statement of the lemma. The statement for $NL_{S/R} \otimes_S S' \to NL_{S'/R'}$ follows as these complexes are homotopic to $NL(\alpha) \otimes_S S'$ and $NL(\alpha')$. □